Machine Learning Project

SR UNIVERSITY

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Batch: 14

Project Title

Rainfall Prediction Model

School of Computer Science and Artificial Intelligence

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DatasetLink:https://www.kaggle.com/datasets/subho117/rainfall-prediction-using-machine-learning?resource=download

GithubRepositoryLink:https://github.com/2403a51357/Rainfall-Prediction-Model

- Dataset used: **Rainfall Dataset**
- Contains weather-related features
- Includes parameters like:
- Temperature
- Pressure
- Humidity
- Cloud
- Wind Direction
- Target variable: **Rainfall (continuous)**

Problem Statement

Predicting rainfall accurately is difficult due to the influence of multiple atmospheric factors such as temperature, humidity, pressure, and wind conditions.
Traditional prediction methods may not capture complex relationships between these variables.
Therefore, a machine learning-based system is required to analyze weather data and provide accurate rainfall prediction.

Explanation of the Project

- Dataset collected and loaded
- Data preprocessing performed:
- Handling missing values
- Encoding categorical data
- Features scaled for better model performance
- Dataset split into training and testing sets
- Three regression models applied:
- Linear Regression
- Decision Tree
- Random Forest
- Models evaluated using RMSE and R^2 score
- Best model selected for rainfall prediction

Algorithms/Models used:

Linear Regression
- Supervised regression algorithm
- Models linear relationship
- Simple baseline model

Decision Tree Regressor
- Tree-based model
- Splits data based on conditions
- Can capture non-linear patterns

Random Forest Regressor
- Ensemble learning method
- Combines multiple decision trees
- Reduces overfitting
- Provides high accuracy

Accuracy / Performance Metrics of the Model

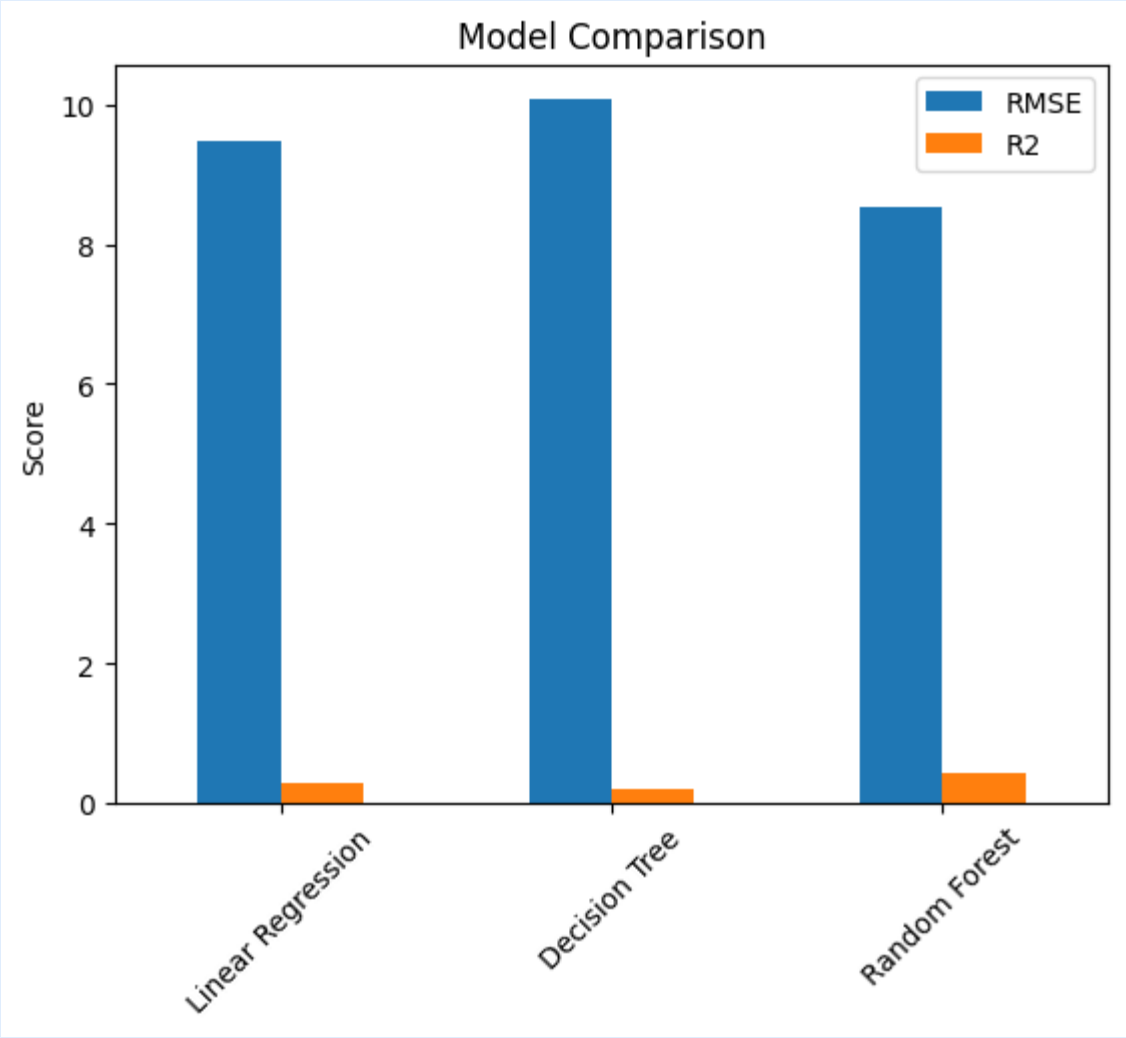
- RMSE (Root Mean Squared Error) measures average prediction error
- Lower RMSE indicates better model performance
- MSE (Mean Squared Error) calculates squared difference between actual and predicted values
- Penalizes large errors more heavily
- R^2 Score measures how well the model explains variance in rainfall data
- R^2 ranges from 0 to 1 (higher is better)
- Linear Regression shows moderate performance due to linear assumptions
- Decision Tree may overfit and give unstable results
- Random Forest achieves lowest RMSE and highest R^2 score
- Random Forest selected as best-performing model
- Metrics confirm model reliability for rainfall prediction

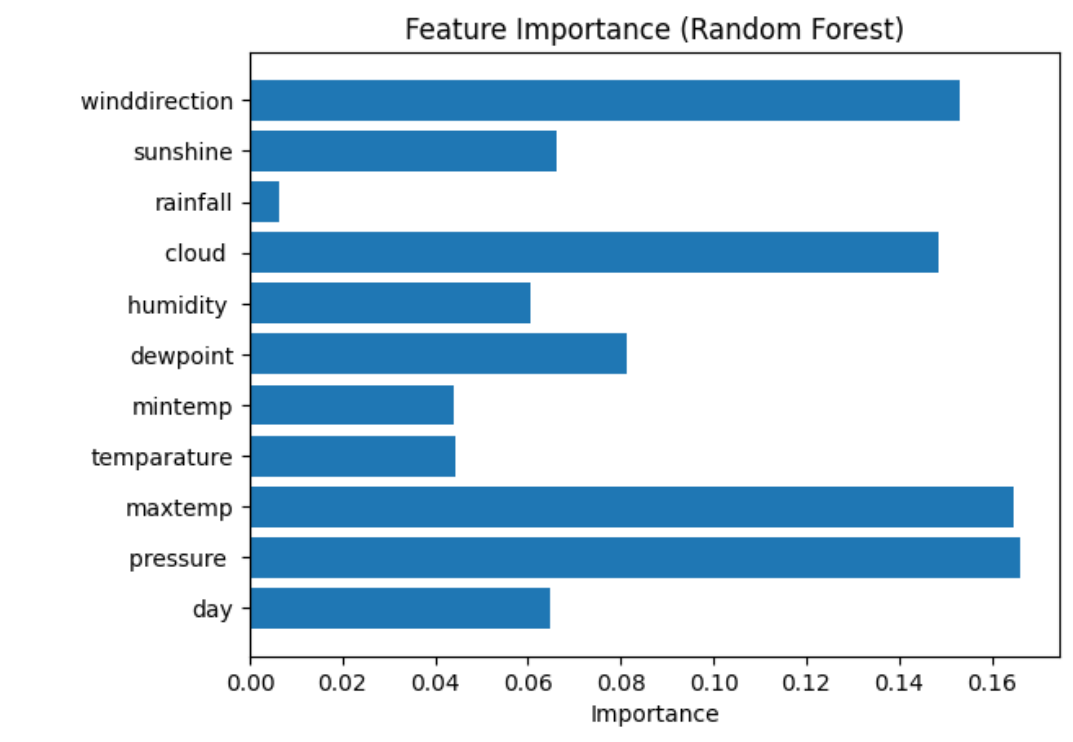
Figures/Plots Showing Results

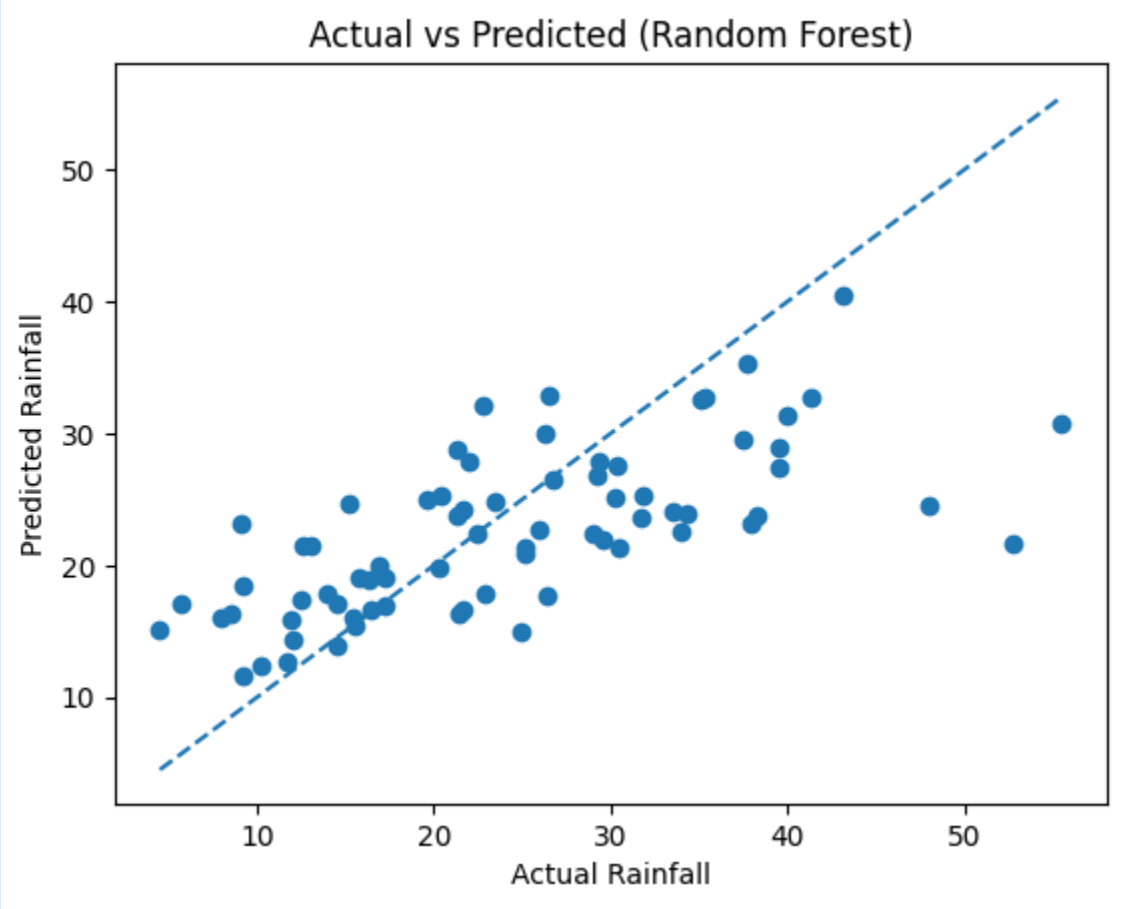
```
Linear Regression
RMSE: 9.478573470038642
R2 Score: 0.29466578151808753

Decision Tree
RMSE: 10.071892260078547
R2 Score: 0.20360021321405608

Random Forest
RMSE: 8.52600333387895
R2 Score: 0.4293103602573294
```




Feature Importance (Random Forest)
winddirection
sunshine
rainfall
cloud
humidity
dewpoint
mintemp
temparature
maxtemp
pressure
day
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Importance


Actual vs Predicted (Random Forest)
Predicted Rainfall
10
20
30
40
50
Actual Rainfall

Outcomes

- Developed a machine learning model for rainfall prediction
- Successfully applied Linear Regression, Decision Tree, and Random Forest models
- Compared models using RMSE and R^2 performance metrics
- Identified Random Forest as the best-performing model
- Achieved improved prediction accuracy using ensemble learning
- Performed data preprocessing including handling missing values and encoding
- Generated visualization graphs such as model comparison and Actual vs Predicted plots
- Analyzed feature importance to understand key factors affecting rainfall
- Built a system capable of predicting rainfall for unseen data
- Demonstrated effectiveness of data-driven approach in weather prediction

Conclusion

- The project successfully developed a machine learning model for rainfall prediction using weather data
- Data preprocessing techniques such as handling missing values and encoding improved dataset quality
- Feature scaling helped enhance model performance and stability
- Three regression models were implemented and compared
- Random Forest achieved the best performance with lowest RMSE and highest R^2 score
- The model effectively captures non-linear relationships in weather parameters
- Visualization graphs validated the accuracy and reliability of predictions
- The final model can be used to predict rainfall for new and unseen data
- This project demonstrates the effectiveness of machine learning in solving real-world weather prediction problems

Methodology / Model Architecture

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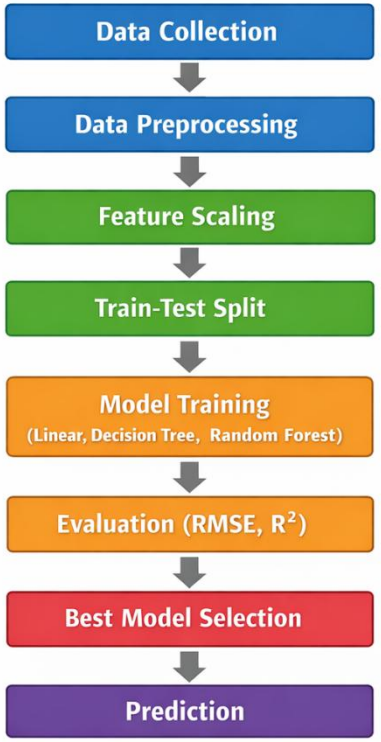
- Flowchart



THANK YOU.